Sidechains and state channels: State channels let people do a lot of transactions off-chain and only write down the final result on the blockchain. This is helpful for things like gaming or micropayments that need a lot of interaction. Sidechains like Polygon are their own blockchains that are linked to Ethereum.

Architecture of a scalable dApp on Layer-2

When you build a scalable decentralised application (dApp) using Layer-2, you need to put together different parts that work together to give users a fast, cheap, and smooth experience. The architecture doesn't just use Layer-1 networks like Ethereum; it also spreads out tasks among frontend, wallet, Layer-2 infrastructure, and backend services. The frontend interface is at the top layer. It is usually built with frameworks like React or just HTML, CSS, and JavaScript. This is where people use the dApp to send transactions, check their balances, or use smart contract features. Next is the integration of the wallet, which is usually done with tools like MetaMask. The wallet links users to the blockchain, securely signs transactions, and lets users switch between Layer-1 and Layer-2 networks.

Choosing the right L2 for your dApp

Choosing the right Layer-2 solution is very important when making a dApp that can grow. You should choose a Layer-2 network based on what your app needs, like speed, cost, security, and user experience. Each one has its own strengths. Polygon chains are a great option if your dApp needs low fees and quick transactions. They are easy to add to and are used by a lot of people, which makes them great for things like gaming, NFTs, and social media. Optimistic rollups like Arbitrum and Optimism are good for apps that need to be very secure and work with Ethereum. They work with current Ethereum smart contracts with only a few changes, which makes it easier for developers to move to the new platform. ZK Rollups like zkSync offer strong cryptographic guarantees and faster withdrawals if you want advanced security and instant finality. However, they may be harder to work with.

Smart contract design considerations on L2

When developers make smart contracts on Layer-2, they need to change how they design them so that they can use all of the scalability features while still being able to work with Layer-1 networks like Ethereum. One important thing to think about is how to save gas. Layer-2 solutions like Polygon and Arbitrum have lower fees, but bad contract code can still cause extra costs. Developers should write code that works as well as possible, use as little storage as possible, and group operations whenever they can.

Cross-layer compatibility is another important thing to think about. Smart contracts should be able to handle deposits, withdrawals, and synchronisation between layers because many dApps work with both Layer-1 and Layer-2. This often means putting bridge contracts together and making sure that data is the same across networks. Even on Layer-2, security is still very important. L2 solutions get some security from Ethereum, but smart contracts can still be hacked. To stop attacks, developers need to follow best practices like checking inputs, controlling access, and doing regular audits.

Data availability and storage strategies

In Layer-2 systems, data availability means how transaction data is stored and accessed so that users can trust and verify the system. Even though execution happens off-chain, the data still needs to be available to make sure networks like Ethereum are safe and open. One common way to make data available on-chain is to post transaction data directly to Layer-1. Arbitrum and other solutions like it follow this model, which makes them very secure because anyone can rebuild and check the state. But it can still be expensive to store data on-chain. Data availability layers are another new strategy.

Bridging and interoperability

Bridging and interoperability are two important parts of scalable dApps that use Layer-2 solutions. Layer-2 networks work with Layer-1 blockchains like Ethereum, so there needs to be a safe way to move data and assets between them. Blockchain bridges make this possible. A bridge lets people move tokens from Layer-1 to Layer-2 to make transactions faster and cheaper. When they need them again, they can bring them back. For instance, a person can send ETH from Ethereum to a Layer-2 network like Polygon to use a dApp for less money. They can later use the same bridge system to get their assets back to Ethereum.

Bridges work by locking up assets on one chain and making the same amount of tokens on the other. When users move back, the tokens are burned on Layer-2 and opened on Layer-1. This makes sure that the total supply stays the same across all networks. Interoperability is more than just moving things around. It lets smart contracts and apps on different Layer-2 networks, like Arbitrum and Optimism, talk to each other and share data. This makes the Web3 ecosystem more connected and adaptable. But bridges need to be built carefully because they can be dangerous if they aren't checked properly. Common problems include delays in withdrawals and possible security holes.

Security considerations

When making scalable dApps with Layer-2 solutions, security is very important. Layer-2 networks get some security guarantees from Layer-1 blockchains like Ethereum, but developers still need to be careful when making apps to avoid security holes at both the smart contract and system levels.